1.1 Propositional Logic - 1

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Definition: A proposition is a declaritive sentence which is true or false (not both).

"Madison is the capitol of Wisconsin." "$x^2+1=2$"

"$1+1=0$" vs. "This statement is false"

"It is 12pm" "What time is it?"

We may discuss one or more propositions and how they relate to each other.

Think of propositions as your boolean phrases.

Definition: A propositional variable is used to denote a proposition. We typically use the letters $p, q, r, s, \ldots$

Since propositions are either true or false, we may explicitly assign a truth value.

If $p =$ "Madison is the capitol of Wisconsin."

then $p \to$ T, $p \to$ F

Definition: The negation of p, denoted by $\neg p, \sim p, -p, Np$ is the statement "It is not the case that p."

If $p =$ "Madison is the capitol of Wisconsin."

then $\sim p =$ "It is not the case that Madison is the capitol of Wisconsin."

* What happened to the truth value of p?
* Is $\sim p$ a proposition?
* How might you say $\sim p$ more naturally?

Truth Table For $\sim$

p	$\sim p$
T	F
F	T

Definition: The conjunction of p and q is denoted by $p \wedge q$, and true only when both p and q are true.

Truth Table For $\wedge$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

The conjunction of at least two propositions is done by inserting the word "and" between them.

If p = "Madison is the capitol of Wisconsin."
q = "$1+1=2$."

then $p \wedge q$ = "Madison is the capitol of Wisconsin <u>and</u> $1+1=2$."

Definition: The disjunction of p and q is denoted by $p \vee q$ and is true if at least p or q is true.

Truth Table For $\wedge$

p	q	$p \wedge q$
T	T	T
T	F	T
F	T	T
F	F	F

The disjunction of at least two propositions is formed by inserting the word "or"

The disjunction of at least two propositions is formed by inserting the word "or" between them.

If p = "Madison is the capital of Wisconsin." and q = "$1+1=2$" then

$p \vee q$ = "Madison is the capital of Wisconsin <u>or</u> $1+1=2$."

<u>Definition</u>: The <u>exclusive or</u> of p and q is denoted by $p \oplus q$ (or p XOR q) and is true only when exactly one of p or q is true.

Truth Table For $\oplus$

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Similar enough to the disjunction. The difference is often that $\oplus$ will include the phrase "but not both." This is sometimes left to context.